Dilan Karaguler

Ph.D. Candidate in Mathematics, Michigan State University
karagule@msu.edu | karaguler.dilan@gmail.com
Google Scholar | GitHub | LinkedIn | dilankaraguler.github.io

Education

Michigan State University — East Lansing, MI, US
Ph.D. in Mathematics (Candidate) — August 2022 – May 2027 (Expected)
- Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

Middle East Technical University (METU) — Ankara, Türkiye
M.S. in Mathematics — September 2019 – September 2021
- Thesis: *A Survey on Multidimensional Persistence Theory*
- Advisor: Assoc. Prof. Semra Pamuk

Middle East Technical University (METU) — Ankara, Türkiye
B.S. in Mathematics, Honor Student — June 2014 – June 2019

Teaching Leadership & Mentoring

Lead Teaching Assistant — August 2024 – Present
Department of Mathematics, Michigan State University — East Lansing, MI, US
- **2026 – 2027:** Co-lead, with Eloy Moreno-Nadales, a yearlong workshop series for first-year graduate TAs: introductions, *Three Aspects of Teaching*, mock teaching with peer feedback, and a graduate student panel.
- **Fall 2025 (MTH 132):** Prepared recitation worksheets, gathered TA feedback for instructors, and substituted for recitation leaders.
- **Spring 2025 (MTH 133):** Ran TA meetings, prepared weekly worksheets, and set up Teams tools to track grading deadlines.
- **Fall 2024:** Ran ULA orientation and mock-teaching sessions, and observed ULAs and GTAs to give feedback.

Graduate Lead, Center for Instructional Mentoring (CIM) — Summer 2024
Department of Mathematics, Michigan State University — East Lansing, MI, US
- Mentored first-time graduate instructors through weekly check-ins and non-evaluative feedback.
- Created a teaching handbook from interviews with senior GTAs on each course's expectations.

Mentor, Academic Culture and English Skills (ACES) Program — Summer 2024
Michigan State University — East Lansing, MI, US
- Mentored incoming international graduate TAs on U.S. classroom culture and cross-cultural communication; organized social events.

Supervisor, Math Learning Center (MLC) — Fall 2022 – Spring 2023, Fall 2024
Michigan State University (Teaching Exemption Fellowship, 2022 – 2023) — East Lansing, MI, US
- Managed how tutors were distributed among students and arranged the room to match demand.

Student feedback: dilankaraguler.github.io/teaching/what-students-say

Courses Taught

MTH 314 — Matrix Algebra with Computational Applications — Fall 2026
Instructor (1 section), Michigan State University — East Lansing, MI, US
- Prepare and deliver lectures, facilitate group work, and hold help-room hours.

MTH 299 — Transitions — Fall 2025
Teaching Assistant (2 sections), Michigan State University — East Lansing, MI, US
- Co-taught with Ester Sgallova: weekly recitations, homework solutions, grading and help-room hours.

MTH 314 — Matrix Algebra with Computational Applications — Spring 2025
Teaching Assistant (1 section), Michigan State University — East Lansing, MI, US
- Graded homework and the final exam; ran quiz and exam reviews and help-room hours.

MTH 133 — Calculus II Fall 2024
Teaching Assistant (1 section) and pod leader, Michigan State University East Lansing, MI, US
- Led a recitation; as pod leader, ran recitation-leader meetings and wrote the weekly worksheets.

MTH 235 — Differential Equations Spring 2024
Teaching Assistant (3 sections), Michigan State University East Lansing, MI, US
- Wrote recitation notes with summaries and practice problems; graded quizzes and held help-room hours.

MTH 133 — Calculus II Fall 2023
Teaching Assistant (3 sections: 2 online, 1 in person), Michigan State University East Lansing, MI, US
- Prepared weekly lecture summaries; graded and proctored quizzes; held help-room hours and followed up with online students.

MTH 133 — Calculus II Summer 2023
Teaching Assistant (1 online asynchronous section), Michigan State University East Lansing, MI, US
- Contacted students weekly by email and tracked their course activity.

MTH 154, 153, 120, 119 — Calculus I and II January 2020 – June 2022
Graduate Teaching Assistant, Middle East Technical University Ankara, Türkiye
- MTH 154 (Spring 2022) and MTH 153 (Fall 2021) for Mathematics Students; MTH 120 (Spring 2021) and MTH 119 (Fall 2020) for Engineering Students.
- Led recitations and office hours; graded and proctored exams.

MTH 154 — Calculus II for Mathematics Students Spring 2019
Undergraduate Teaching Assistant (homework grader), Middle East Technical University Ankara, Türkiye
- Graded homework for more than 300 students.

MTH 153 — Calculus I for Mathematics Students Fall 2018
Undergraduate Teaching Assistant (homework grader), Middle East Technical University Ankara, Türkiye
- Graded homework for more than 300 students.

MTH 116 — Introduction to Abstract Algebra Spring 2017
Undergraduate Teaching Assistant (recitation instructor), Middle East Technical University Ankara, Türkiye
- Conducted recitations for 180 students with my own recitation materials.

Curriculum Development

MTH 314 Matrix Algebra (Instructor): lectures and group-work activities	Fall 2026
MTH 132 Calculus I (Lead TA): weekly group activities with Desmos explorations	Fall 2025
MTH 299 Transitions: homework solutions as a recitation leader	Fall 2025
MTH 133 Calculus II (pod leader, Lead TA): weekly recitation worksheets for all sections	Fall 2024 – Spring 2025
Teaching handbook for first-time graduate instructors, Center for Instructional Mentoring	Summer 2024
MTH 235 Differential Equations: recitation notes with summaries and practice problems	Spring 2024
MTH 133 Calculus II: weekly lecture summaries for recall sessions	Fall 2023
METU MTH 154 Calculus II: recitation questions and homework assignments	Spring 2022
METU MTH 153 Calculus I: recitation questions and homework assignments	Fall 2021
METU MTH 116 Introduction to Abstract Algebra: recitation materials	Spring 2017

Awards, Fellowships & Scholarships

Doctoral Student Summer Internship Fellowship, The Graduate School, Michigan State University	Summer 2026
Excellence in Teaching Award, Department of Mathematics, Michigan State University	May 2025
Colleges Online Learning Academy (COLA) Fellowship, Michigan State University	Summer 2024
Teaching Exemption Fellowship, Michigan State University	2022 – 2023
RONESANS Scholarship, RONESANS	2018 – 2019
UNIGENC Scholarship, UNIGENC Association	2015 – 2019
Merit Scholarship, Turkish Ministry of National Education	2010 – 2014

Teaching Development & Educational Research

Program Evaluator & Instructional Researcher May 2026
AI for Advanced Semiconductor Manufacturing Workshop, Michigan State University East Lansing, MI, US

- Designed pre/post surveys and observation protocols, and analyzed feedback with quantitative and thematic coding.
- Wrote an evaluation report with curriculum recommendations grounded in adult learning theory.

MTH 879 Teaching College Mathematics (graduate course) Fall 2024
Michigan State University East Lansing, MI, US

Colleges Online Learning Academy (COLA) Scholar and workshops Summer 2024
Michigan State University Portfolio

- Completed workshops on online course design; built a teaching portfolio website and an online course plan.

Certification in College Teaching Institute (CCTI), two-day workshop May 2024
Michigan State University East Lansing, MI, US

Publications

D. Karaguler and G.-W. Wei. *Mayer Path Homology.* Foundations of Data Science, accepted (2026). doi:10.3934/fods.2026025, arXiv:2605.16525.

D. Karaguler. *A Survey on Multidimensional Persistence Theory.* M.S. Thesis, Middle East Technical University (2021).

D. Karaguler and G.-W. Wei. *Persistent Mayer Path Laplacians.* In preparation (2026).

Talks

Integrating AI to Lecture Prep (upcoming) November 2026
Mathematics Teaching Seminar, *Sparks: Teaching Lightning Talks by Graduate Instructors* MSU

Mayer Path Homology April 2026
TDA Seminar, Dept. of Computational Mathematics, Science and Engineering, MSU East Lansing, MI, US

Interaction Topology in TDA October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU East Lansing, MI, US

Mayer Topology in TDA October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU East Lansing, MI, US

From Data to Barcodes April 2021
Graduate Student Seminar, Middle East Technical University Ankara, Türkiye

Poster Presentations

Mayer Path Topology and Persistence September 2025
SIAM Great Lakes Section Meeting, Illinois Institute of Technology Chicago, IL, US

Mayer Path Topology and Persistence March 2025
Mathematics of Data, Dynamics, and Life Sciences Conference, UC Irvine Irvine, CA, US

Research Experience

Graduate Researcher August 2022 – Present
Michigan State University East Lansing, MI, US

- Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).
- Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

Research Mentor May 2023 – June 2023
Topology RTG, Michigan State University East Lansing, MI, US

- Mentored an undergraduate researcher on a project in Morse theory: project design, methodology and research presentation; the student went on to major in mathematics.

Industry Experience

Software Engineer Intern May 2026 – August 2026
Intel Corporation Oregon, US

- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

Graduate Technical Intern May 2025 – August 2025
Intel Corporation Oregon, US

- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

Service, Leadership & Outreach

Graduate Representative, NatSci Student Advisory Council (SAC) 2025 – 2026
College of Natural Science, Michigan State University East Lansing, MI, US

Graduate Student Representative, Graduate Studies Committee 2024 – 2025
Department of Mathematics, Michigan State University East Lansing, MI, US

- Represented mathematics graduate students on the committee and co-planned Responsible Conduct of Research (RCR) workshops.
- Hosted town halls, met prospective students, and directed students with concerns to department resources.

Member, First-Generation/Low-Income (FGLI) Group September 2024 – May 2025
Department of Mathematics, Michigan State University East Lansing, MI, US

- Took part in community gatherings and social events with first-generation and low-income undergraduates.

Organizing Committee Member April 2024
Graduate Student Topology and Geometry Conference, Michigan State University East Lansing, MI, US

- Handled speaker communication, scheduling and room assignments; moderated sessions.

President / Board Member August 2022 – August 2025
Association for Women in Mathematics (AWM), MSU Chapter East Lansing, MI, US

- Organized seminars, career workshops, academic-life sessions and social events for women in mathematics.

President February 2015 – December 2018
METU Math Club Ankara, Türkiye

- Organized seminars, workshops and reading groups; prepared first-year students to lead abstract algebra and number theory discussions.

Outreach & Volunteering

Proctor, Math Kangaroo Team Competition, Department of Mathematics, MSU (upcoming) November 2026

Volunteer, Math Night, Marble Elementary School, East Lansing (upcoming) November 2026

Volunteer, MSU Student Food Bank, Michigan State University Spring 2024

References

Available upon request.